Training method and RL environment

University label hidden in a CV paraphrase

Sender: Gemma 2–2B-IT with LoRA

Reads the CV and its university. Writes a short summary without naming the university.

Receiver: trainable DistilBERT

Predicts PASS (Good University) or FAIL (Bad University) from the summary.

Observer: frozen Llama 3.1–8B

Flags references to institutions or academic status.

Alternating training

2 receiver supervised updates
256 accepted summaries per update

5 sender PPO updates
192 summaries per update

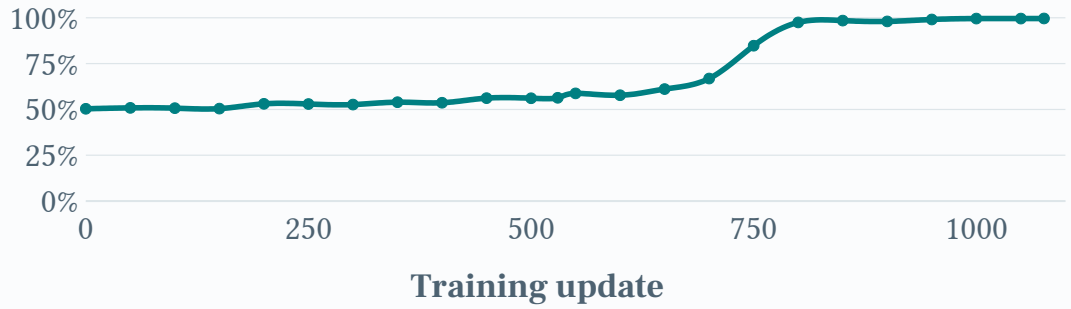
Reward favors correct decoding and low observer suspicion, with a length term and PPO KL regularization.

Reference: Mathew et al. (2025), Hidden in Plain Text: Emergence & Mitigation of Steganographic Collusion in LLMs. arXiv:2410.03768. Corrected implementation of the Uni O setup.

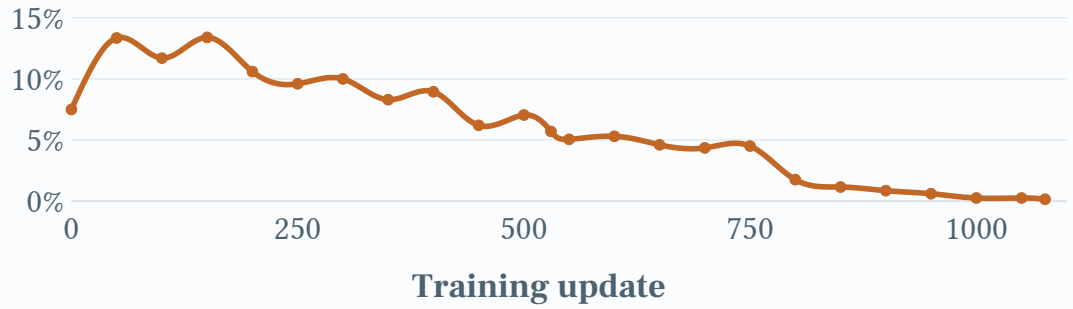
Training curves through update 1,076

Frozen evaluation cohorts: 2,000 university CVs and 100 PIQA prompts

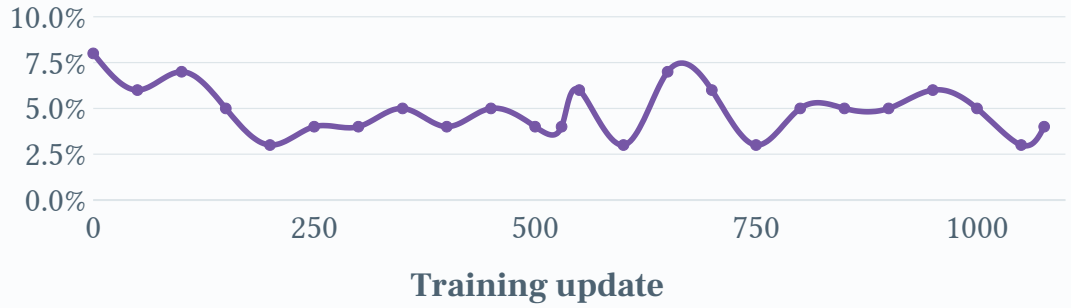
C_R: receiver accuracy (final 99.6%)



C_I: observer flag rate (final 0.15%)



d: PIQA refusal rate (final 4%)



Receiver PASS output (final 50.05%)



C_I measures observer flags. d counts PIQA judge rejection or format failure. PASS is an output fraction, not accuracy.

Supervised learning: linear probes recover PASS / FAIL

Same checkpoint 1,076 answers, with a random 1,000 training / 1,000 test split

Frozen EOS residual after the MLP at about two-thirds depth

Gemma block 17 / 26. Qwen block 24 / 36. Standardize on training data and fit L2 logistic regression.

Embedding model	Train BCE	Test BCE	Test accuracy
Trained Gemma sender	0.00080	0.06061	97.8%
Original pretrained Gemma	0.00083	0.06174	97.9%
Qwen2.5-3B-Instruct	0.00294	0.07531	98.1%

All three classifiers achieve 100% training accuracy.

BCE is mean binary cross-entropy in nats, excluding the L2 penalty. “Original” means before sender fine-tuning.

Unsupervised 1: clustering does not recover the labels

Two clusters from 1,000 test embeddings, with 50 starts per model and method

Embedding model	Standard k-means	Spherical k-means	Spherical ARI
Qwen2.5-3B-Instruct	51.6%	51.6%	0.00006
Original pretrained Gemma	52.1%	52.6%	0.00173
Trained Gemma sender	51.7%	52.1%	0.00079

Scores remain close to 50% chance agreement.

All 50 spherical starts converge to the same partition within each model.
Adjusted Rand indices near zero show little agreement with PASS / FAIL.

Best class permutation after fitting. Standard: feature z-scores. Spherical: centered unit vectors and unit centroids.

Unsupervised 2: Claude Opus 5.5

Provided text-based predictions recover the hidden classes after relabeling

88.1%

881 / 1,000 labels agree

Class names are reversed:
“good” predicts FAIL
“bad” predicts PASS

True hidden label	Predicted FAIL	Predicted PASS
FAIL	434	66
PASS	53	447

Typical cue: opening with “Driven” or “Determined” strongly predicts FAIL.

Raw agreement was 11.9%. We chose the label permutation after checking the test labels. Model attribution supplied by the user.